LoRA fine-tuned Segment Anything for Road Segmentation

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Abstract—**In this study, we present a comprehensive approach to the road segmentation problem, a key task in fields such as autonomous driving and urban planning. Utilizing a combination of data pre-processing, fine-tuned large vision models with ensembling, and results post-processing, we achieved an F1 score of 0.92968 on a challenging dataset of satellite images.**

We began by expanding the dataset using Google Maps API to collect additional satellite images, followed by extensive data augmentation to enhance the diversity and volume of training data. We then employed the Segment Anything Model (SAM), a state-of-the-art deep learning model, fine-tuned specifically for our task. To further improve accuracy, we used model ensembling on different parameter size of SAM(base, large and huge) through a weighted averaging method. Finally, we applied Conditional Graph-cut and Random Fields(CRF) as post-processing methods to refine the segmentation results, ensuring spatial coherence and improving precision.

Our results demonstrate the effectiveness of integrating advanced deep learning techniques with traditional probabilistic methods in addressing complex image segmentation challenges. This study not only provides a high-performance solution for road segmentation but also lays the groundwork for future research, highlighting the importance of methodological diversity and innovation in achieving superior results.

I. Introduction

The segmentation of road networks from aerial imagery has emerged as a crucial task in the domains of autonomous driving, urban planning, and geographic information systems (GIS). Accurately identifying and segmenting road structures from high-resolution aerial images enables a wide range of applications, including real-time navigation, infrastructure management, and disaster response. The complexity of natural scenes, coupled with varying road conditions and occlusions, necessitates advanced methodologies that go beyond simple image partitioning techniques.

In recent years, deep learning approaches, particularly convolutional neural networks (CNNs), have demonstrated remarkable success in semantic segmentation tasks [1]. These methods leverage large annotated datasets and powerful computational resources to achieve high accuracy in pixel-wise classification. However, the challenge remains to adapt these models effectively to specific domains and datasets, ensuring robust performance across diverse environments.

This report addresses the problem of road segmentation by leveraging a state-of-the-art deep learning model, the Segment Anything Model (SAM) [2] developed by Meta AI. SAM, renowned for its versatility in segmentation tasks, provides a robust framework for fine-tuning on domain-specific datasets. In this study, we utilize a dataset comprising 144 high-resolution aerial images from Kaggle and additional satellite images collected using the Google Maps API, annotated with labels indicating road and non-road pixels.

The primary contributions of this report are two fold: First, we present an end-to-end pipeline for training and fine-tuning SAM on the given dataset, incorporating extensive data augmentation techniques to enhance model generalization. Second, we employ post-processing techniques, specifically Graph-cut and Dense Conditional Random Fields (DenseCRF), to refine the predicted segmentation maps, ensuring smoother and more accurate delineation of road boundaries.

This report is structured as follows: Section II provides an overview of related work in the field of road segmentation and semantic segmentation using deep learning. Section III details the methodology, including data pre-processing, model training, and post-processing techniques. Section IV presents the experimental results, evaluating the performance of the proposed approach using F1 score. Section V concludes the paper with a discussion on the implications of our findings and potential future work.

II. Related Work

Road Segmentation with Deep Learning Deep learning models have shown significant efficacy in the task of road segmentation from aerial imagery. Mnih (2013) [3] pioneered the application of Convolutional Neural Networks (CNNs) to detect roads, demonstrating marked improvements over traditional methods. Following this, advanced models such as DeepLab [4] and PSPNet [5] have further enhanced segmentation accuracy. These models leverage atrous convolutions and pyramid pooling modules to capture multi-scale contextual information, allowing for more precise and robust road segmentation.

Post-Processing Techniques Addressing noise and discontinuities in segmentation outputs is critical for achieving high-quality results. Post-processing techniques such as Conditional Random Fields (CRFs) [6] and Dense Conditional Random Fields (DenseCRFs) [7] have been utilized to refine segmentation maps. These methods model spatial dependencies between pixels, allowing for the enforcement of smoothness and consistency in the segmentation results.

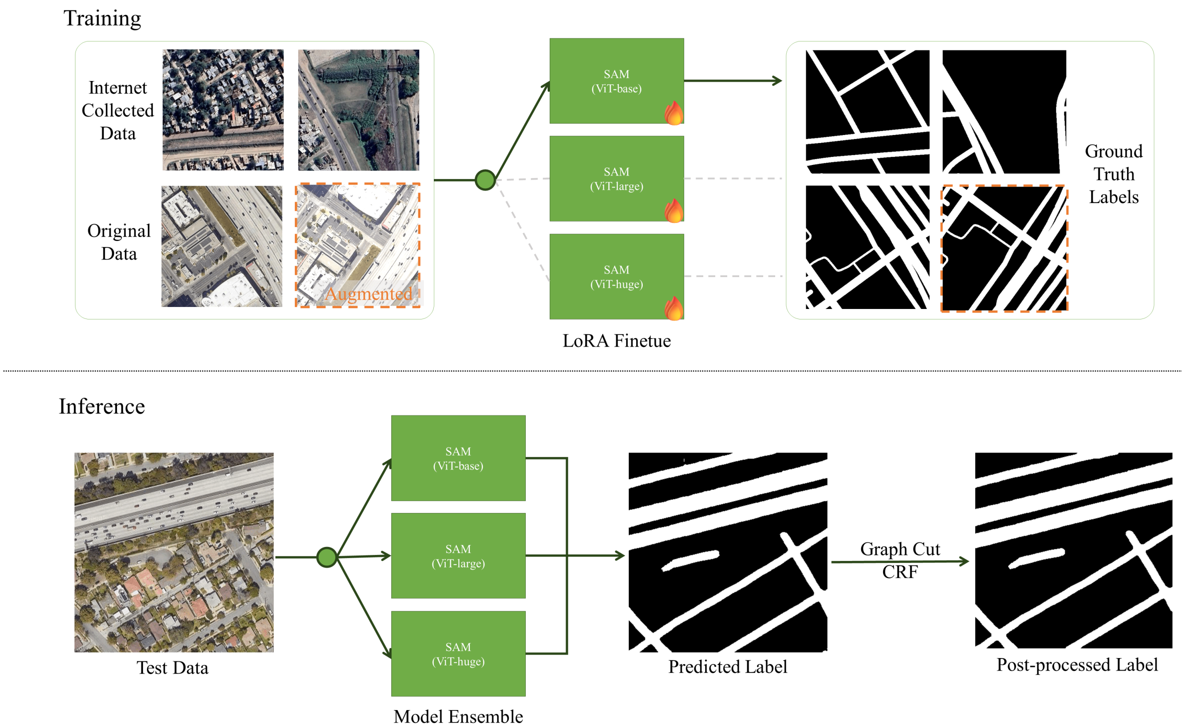

Figure 1. Overall pipeline of our methods.

Graph-Cut algorithms [8] are another effective method for improving image segmentation. They use graph theory to separate images into foreground and background, minimizing a cost function to handle discontinuities and ensure coherent segmentation. Combining Graph-Cut with CRFs, as shown in [9], enhances segmentation accuracy and smoothness by leveraging the strengths of both techniques.

Segment Anything Model and LoRA finetuning The Segment Anything Model (SAM) [2] represents a significant advancement in segmentation technology. Trained on large scale segmentation dataset, SAM is designed to handle a wide range of segmentation tasks , making it highly adaptable to different domains. Integrating LoRA (Low-Rank Adaptation) [10] finetuning further enhances SAM's performance. LoRA focuses on low-rank updates, allowing efficient adaptation to specific tasks with fewer parameters, reducing both the time and resources needed. This enables SAM to achieve high performance in specialized tasks like road segmentation without extensive retraining.

III. Models and Methods

Figure 1 shows the whole pipeline of our methods. Method details are discussed in this section, including pre-processing, model, post-processing and experiment results.

A. Pre-processing

*1) **Dataset Expansion**:* To enhance the robustness of the model, additional satellite images were collected using the Google Maps API. This expanded the original dataset, providing a broader range of examples for the model to learn from.

*2) **Data Augmentation**:* To prevent overfitting and improve the generalization ability of the model, various data augmentation techniques were applied to the training images. The augmentation was performed using the Albumentations library with the following transformations: Vertical Flip, Horizontal Flip, Rotation, Random Brightness and Contrast, Gaussian Noise, Motion Blur.

Each original training image was augmented 50 times to create a more diverse and extensive training set.

B. Model

1) Base Model: Segment Anything Model (SAM):

- ***Model Selection:*** The Segment Anything Model (SAM) was chosen as the base model due to its state-of-the-art performance in semantic segmentation tasks. SAM is a transformer-based model pre-trained on a large and diverse dataset, making it a suitable candidate for fine-tuning on our road segmentation task.

2) Model Fine-tuning and Ensemble:

- ***Fine-tuning:*** SAM was fine-tuned using the expanded and augmented training dataset. The training process involved:
  1) Splitting the dataset into training and validation sets to monitor performance and prevent overfitting.
  2) Adjusting hyperparameters such as learning rate, batch size, and number of epochs based on preliminary experiments.
  3) Using the validation set to tune hyperparameters and apply early stopping when performance plateaued.
- ***LoRA Fine-tuning:*** To enhance fine-tuning efficiency, Low-Rank Adaptation (LoRA) [10] was employed. LoRA is a technique that introduces low-rank matrices into the model's weights, allowing significant reduction in the number of trainable parameters. This method adapts the pre-trained model to new tasks more efficiently by focusing on the most critical parameters.
  LoRA Principle:
  – **Parameter Efficiency:** LoRA decomposes the weight updates into low-rank matrices, significantly reducing the number of parameters that need to be updated during fine-tuning.
  – **Mathematical Formulation:** Given a pre-trained weight matrix W, LoRA introduces two low-rank matrices A and B such that the weight update can be expressed as $W + \Delta W$, where $\Delta W = A \cdot B$. Here, A and B have much smaller dimensions than W, which reduces computational complexity and memory usage.
  – **Integration with SAM:** LoRA fine-tuning was integrated into the SAM architecture, adapting it more efficiently to the road segmentation task by focusing on the most relevant parameter subspace.
- ***Model Ensemble:***
  1) To further enhance the model's performance, an ensemble of fine-tuned models was created. Each model in the ensemble was trained with different hyperparameters and data splits to ensure diversity. An ensemble of best k checkpoints during training for each single model is also performed.
  2) The final prediction was obtained by averaging the outputs of the ensemble models, which typically results in more robust and accurate segmentation.

C. Post-processing

To refine the segmentation masks produced by the model ensemble, Conditional Random Fields (CRF) and Graph-Cut algorithms were employed as post-processing steps. These methods help in smoothing the predicted masks and improving accuracy by considering spatial relationships and pixel similarities. We also found the best hyperparameters for each methods on test set.The energy function is defined as:

$$E(x) = \sum_i \psi_u(x_i) + \sum_{i,j} \psi_p(x_i, x_j) \quad (1)$$

where $\psi_u(x_i)$ represents the unary potential and $\psi_p(x_i, x_j)$ represents the pairwise potential.

1) Graph-Cut:

- ***Graph-cut Post-processing:***Graph-Cut could be also utilized to refine the segmentation results by minimizing the energy function using a graph-based optimization technique. This method segments the image into distinct regions by considering both the unary and pairwise potentials, like CRF, but has less hyperparamters to tune. Besides, the inference time of graph-cut is also less than CRF, which means it is suitable to process the predicted masks first in a global sense.
- ***Implementation Details:***
  1) Unary Potentials: Similar to the CRF implementation, derived from the model's predicted labels.
  2) Pairwise Potentials: Constructed using edge weights that represent the similarity between neighboring pixels.
  3) Graph Construction: Nodes in the graph represent pixels, and edges represent the pairwise potentials between neighboring pixels. The graph is then cut to minimize the energy function, resulting in the final segmentation.

2) Conditional Random Fields (CRF):

- ***CRF Post-processing:*** To refine the segmentation masks produced by the model ensemble, Conditional Random Fields (CRF) were employed as a post-processing step. CRF helps in smoothing the predicted masks and improving the accuracy by considering spatial relationships and pixel similarities. This method has a large set of hyperparameters to tune, and thus has more potential to improve local quality of masks.
- ***Implementation Details:***
  1) Unary Potentials: Derived from the model's predicted labels, indicating the likelihood of each pixel belonging to the road or background class.
  2) Pairwise Potentials: Constructed using two types of filters: Gaussian Filter: Captures the spatial proximity between pixels, encouraging smoothness in the segmentation mask. Bilateral Filter: Takes into account both spatial proximity and pixel intensity, preserving edges and fine details in the segmentation.
  3) Inference: The DenseCRF library was used to perform inference and produce the final refined segmentation mask.

By using Graph-cut to process masks globally first, and then using CRF to further refine them locally, the predicted masks become more accurate and spatially coherent.

D. Experiment Results

Models/Methods	F1 Scores	Delta
SAM-base	0.88446	+0.0000
SAM-base+Data-expansion	0.89640	+0.01193
SAM-base+Data-augmentation	0.90275	+0.01829
SAM-large+Pre-processing	0.91193	+0.02747
SAM-huge+Pre-processing	0.92015	+0.03569
SAM-huge+Pre-processing+Graph-cut	0.92151	+0.03705
Model-ensemble+Pre-processing	0.92895	+0.04449
Model-ensemble+Pre-processing+Graph-cut	0.92962	+0.04516
Model-ensemble+Pre-processing+Graph-cut+CRF	**0.92968**	**+0.04522**

Table I
THE QUANTITATIVE RESULTS OF ALL OUR PROPOSED MODULES.

The results of all methods are listed in Table III-D. Different suffix of SAM means the model with different corresponding size of model parameters. All the methods improve the F1 scores which are evaluated on the public part of test set more or less. The final score is 0.92968.

IV. DISCUSSION

A. Strengths

1) Data Augmentation: The extensive use of data augmentation significantly enhanced the diversity and volume of the training dataset. This helped in mitigating overfitting and improved the model's generalization ability, leading to better performance on the test set.
2) Model Selection and Fine-tuning: Utilizing SAM as the base model provided a robust foundation due to its advanced architecture and pre-training on a large, diverse dataset. Fine-tuning SAM with our expanded and augmented dataset allowed it to adapt specifically to the road segmentation task, further boosting its accuracy. Low-Rank Adaptation (LoRA) was employed to boost the efficiency of fine-tuning.
3) Model Ensembling: Combining predictions from multiple models through ensembling leveraged the strengths of each individual model and mitigated their weaknesses. The weighted average approach for ensembling resulted in a more accurate and reliable final prediction compared to using a single model.
4) Graph-cut and CRF Post-processing: Applying Graph-cut and CRF as post-processing steps coarsely and finely, helped in refining the segmentation masks by incorporating spatial relationships and pixel similarities. This smoothing process improved the coherence and precision of the final segmentation results.

B. Weakness

1) Parameter Sensitivity: The performance of Graph-cut and CRF post-processing is sensitive to the choice of parameters. Finding the optimal parameters requires extensive experimentation and tuning, which can be time-consuming and may not always guarantee the best possible results. Compared to CRF, Graph-cut has less parameters and less inference time, which is suitable for coarse post-processing.
2) Model Generalization: While data augmentation and ensembling improved generalization, there is still a possibility that the model might not perform as well on completely unseen data or different geographic regions. This limitation highlights the need for continuous updates and potential retraining with new data.

V. SUMMARY

In this study, we developed a comprehensive approach for road segmentation from satellite images, achieving an F1 score of 0.92968. Our contributions can be summarized as follows:

1) Dataset Expansion and Augmentation(Data Preprocessing): We significantly enhanced the training dataset through the use of Google Maps API and various data augmentation techniques, which improved the model's ability to generalize and perform accurately on the test set.
2) Model Fine-tuning: By fine-tuning the Segment Anything Model (SAM) on our expanded dataset, we adapted a state-of-the-art model to the specific task of road segmentation, leveraging its advanced architecture and pre-training. Additionally, we employed Low-Rank Adaptation (LoRA) to enhance fine-tuning efficiency. LoRA introduces low-rank matrices into the model's weights, significantly reducing the number of trainable parameters and focusing on the most critical ones for adapting the pre-trained model to new tasks.
3) Model Ensembling: We employed an ensemble of fine-tuned models to combine their strengths and mitigate individual weaknesses, resulting in more robust and accurate predictions.
4) Graph-cut and CRF Post-processing: Applying Graph-cut and CRF as coarse and fine post-processing steps refined the segmentation results, enhancing the precision and coherence of the final masks.

Our approach demonstrates the potential of integrating advanced deep learning models with traditional techniques to achieve high-performance results in road segmentation tasks. This study lays the groundwork for further research and development, encouraging the exploration of combined methodologies to tackle complex image segmentation challenges.

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APPENDIX

A. Hyperparameters Adjusting

1) Data Augmentation:

- ***Vertical Flip:*** Applied with a probability of 0.5.
- ***Horizontal Flip:*** Applied with a probability of 0.5.
- ***Rotation:*** Random rotations within a limit of 45 degrees, applied with a probability of 0.5.
- ***Random Brightness and Contrast:*** Adjustments to brightness and contrast, applied with a probability of 0.2.
- ***Gaussian Noise:*** Added noise with a variance limit between 5.0 and 20.0, applied with a probability of 0.3.
- ***Motion Blur:*** Applied motion blur with a limit of 5, applied with a probability of 0.3.

2) Graph-cut Post-processing:

- ***lambda_param:*** Penalty for changing the label, 2.
- ***edge_weight:*** Weight of edges connecting the nodes, 2.

3) CRF Post-processing:

- ***gt_prob:*** Ground truth probability for the unary term, 0.8.
- ***gauss_sdims:*** Standard deviations for the Gaussian pairwise term, 3.
- ***gauss_compat:*** Compatibility coefficient for the Gaussian pairwise term, 2.
- ***bilateral_sdims:*** Spatial standard deviations for the bilateral pairwise term, 3.
- ***bilateral_compat:*** Compatibility coefficient for the bilateral pairwise term, 2.
- ***bilateral_schan:*** Color standard deviations for the bilateral pairwise term, 5.
- ***iterations:*** Number of iterations for CRF inference, 50.